PRACTICAL APPLICATION:

Security Metrics and Executive Reporting:

Metrics Dashboard:

Creating Elastic Security dashboard for MTTD, MTTR, and false positive rate,

1) start ELK ,

```
sudo systemctl start elasticsearch
sudo systemctl start kibana
sudo systemctl start logstash
sudo systemctl start rsyslog
```

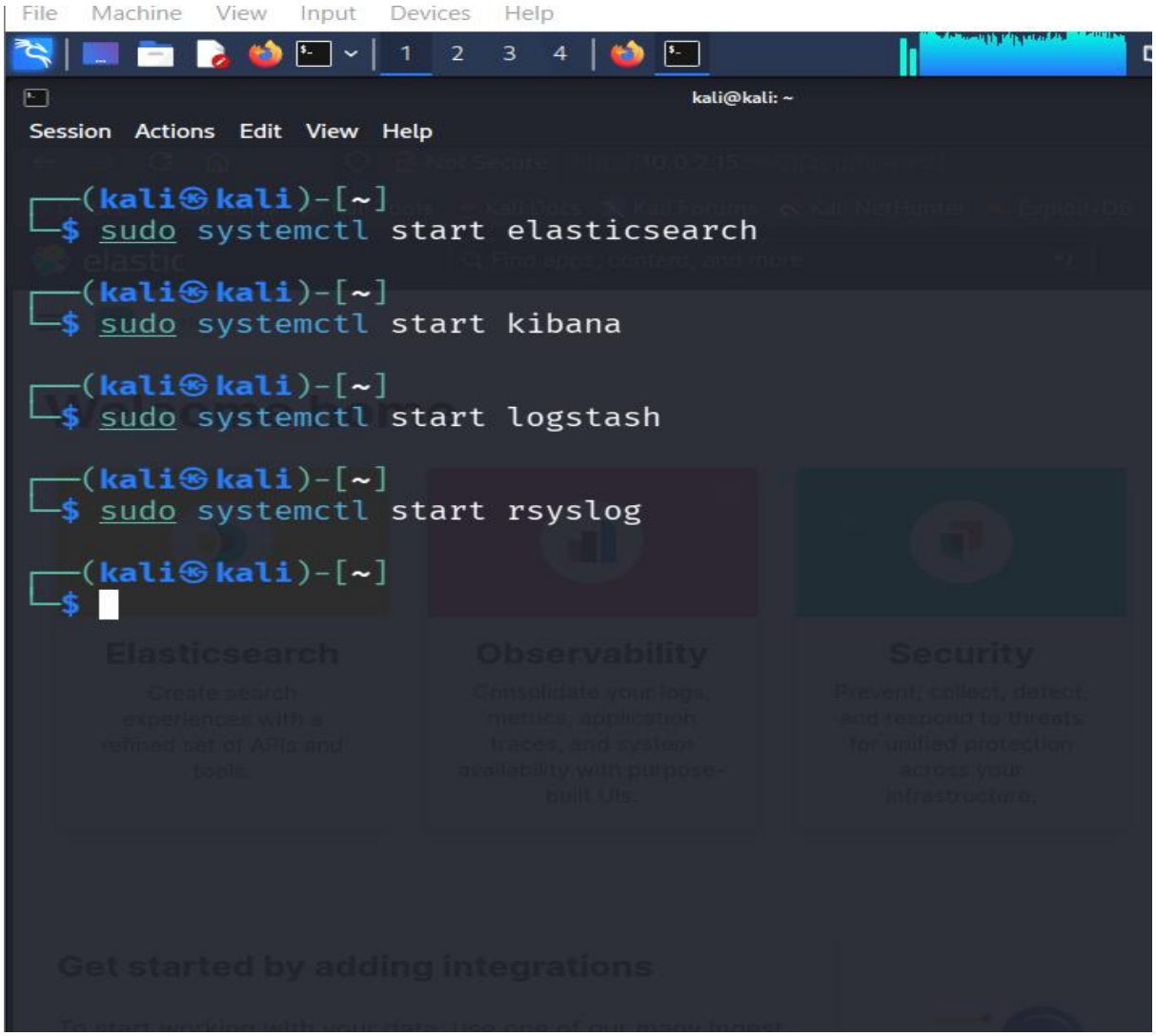

Figure 1: starting ELK

Then access elasticsearch through browser,

http://10.0.2.15:5601

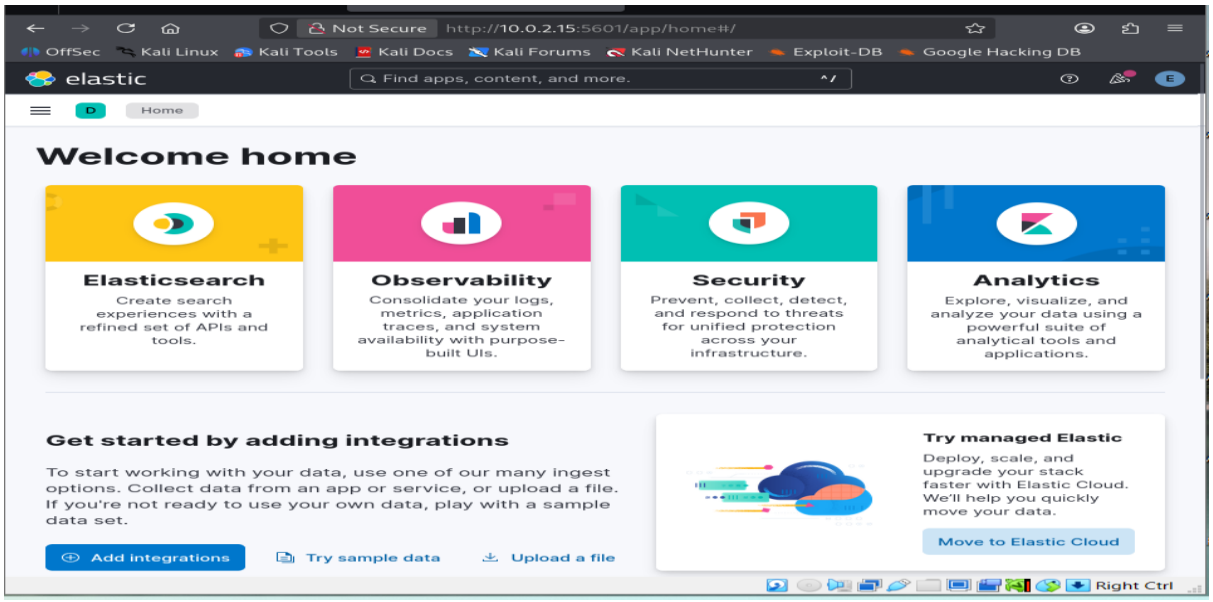

Figure 2: accessing elastic through browser

2) add mock incident data:

Create a csv file and upload it to elastic,

sudo nano soc_metrics.csv

and add below shown screenshot data to the file

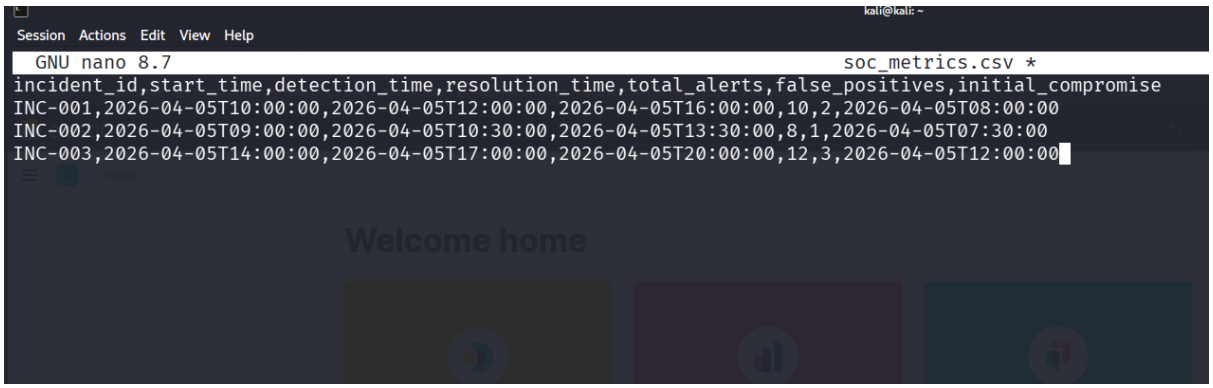

Figure 3: adding mock incident data to the csv file

3) upload csv file into kibana,

Go to Analytics → Click Machine Learning→ data visualizer →Select Upload file → Upload soc_metrics.csv → Name the index something like:

soc-metrics

Finish import

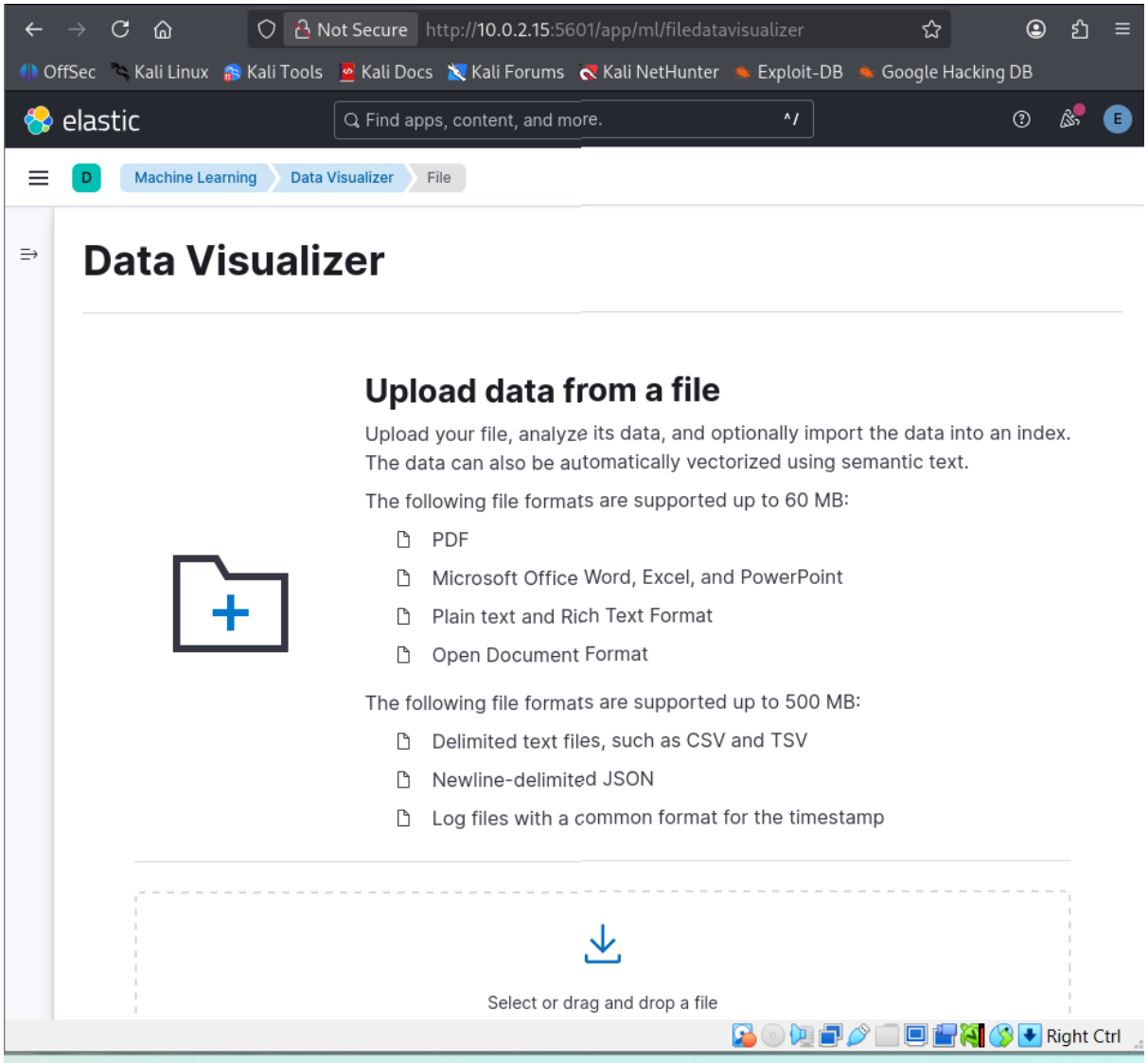

Figure 4: data visualizer, here upload the file

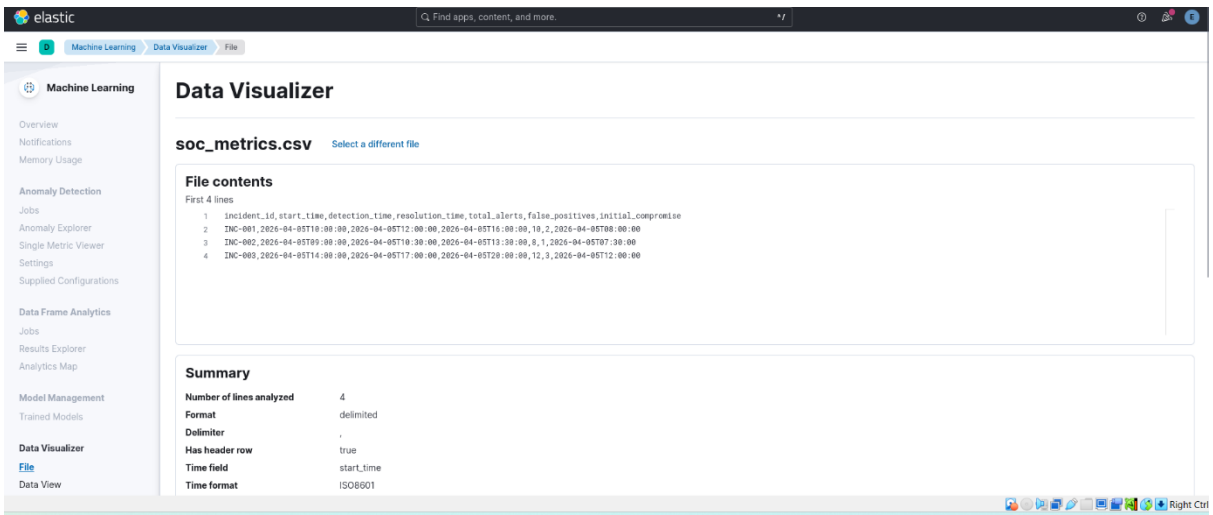

Figure 5: file contents after uploading

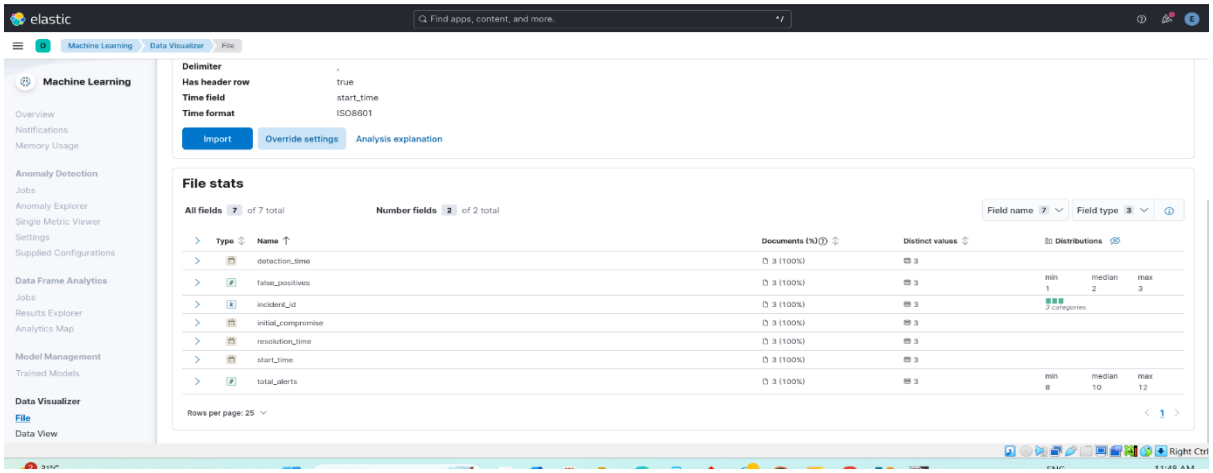

Figure 6: file contents after uploading

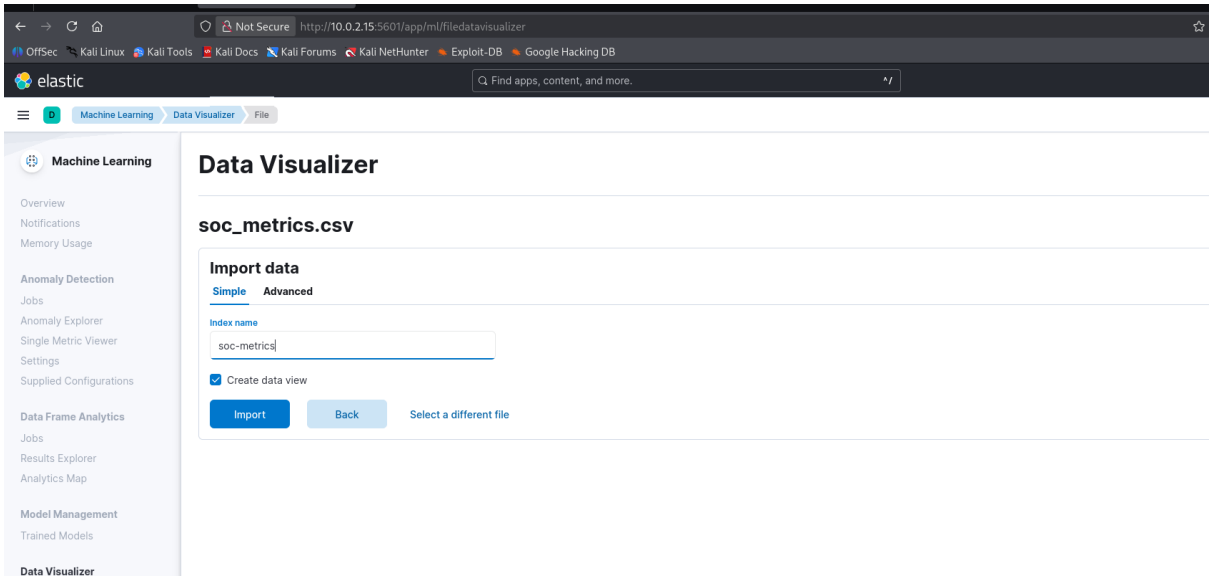

Figure 7: name index and import

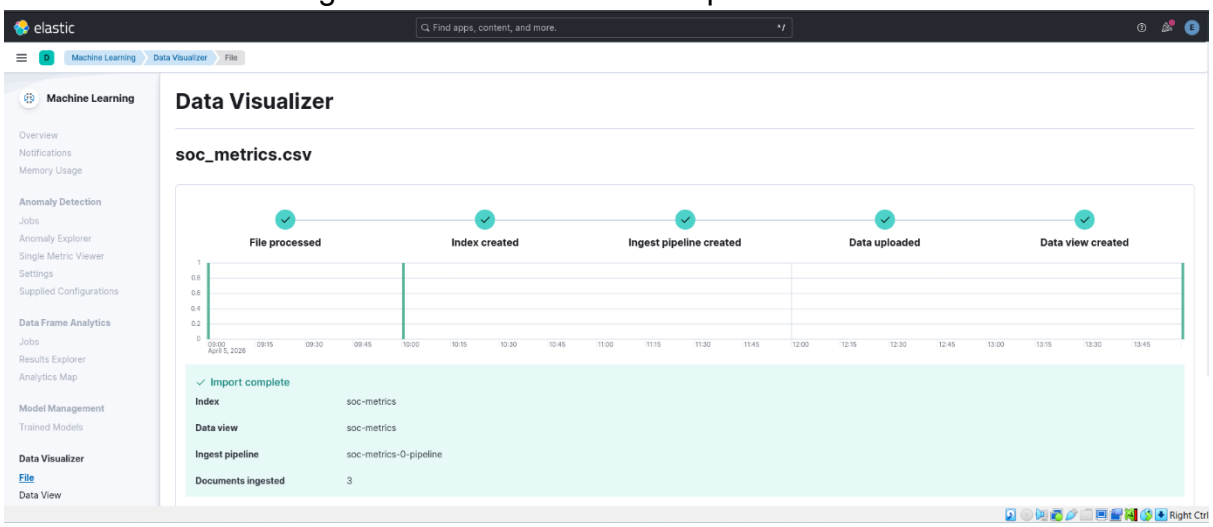

Figure 8: data is imported successfully and now data is inside elastic

4) Create Calculated Fields in Kibana:

Create fields for MTTD, MTTR, Dwell time, False positive rate.

Goto stack management → dataviews → open soc-metrics datavies → click add field.

Field 1 – MTTR :

Field name: mttd_hours

Type: double

Then goto set value and add script and save it

Script:

```
if (doc['detection_time'].size() != 0 && doc['start_time'].size() != 0) {
        emit((doc['detection_time'].value.toInstant().toEpochMilli() -
doc['start_time'].value.toInstant().toEpochMilli()) / 3600000.0);
}
```

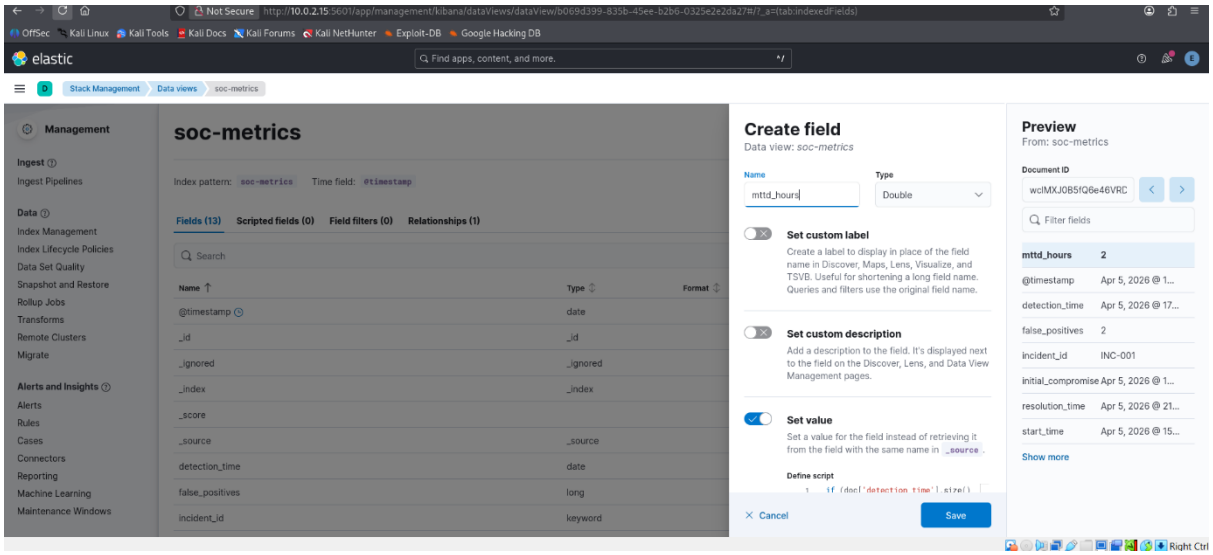

Figure 9: creating mttd_hours field

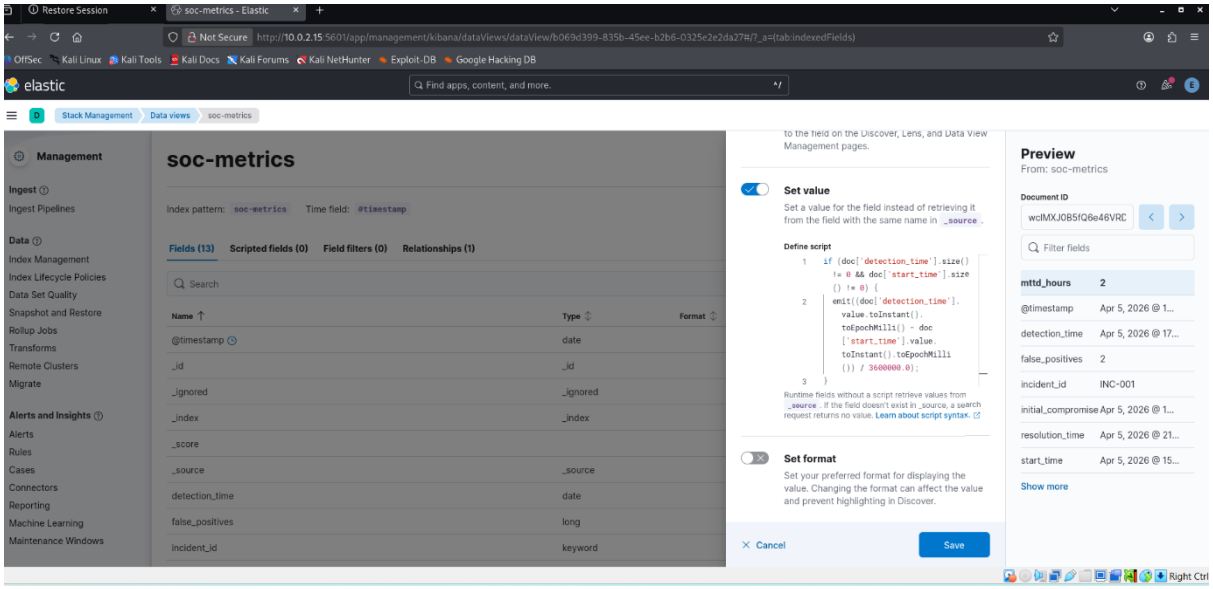

Figure 10: adding script

It calculates MTTD and Then converts it into hours.

Field 2 – MTTR:

Name: mttr_hours

Type: double

Script:

```
if (doc['resolution_time'].size() != 0 && doc['detection_time'].size() != 0)
{
  emit((doc['resolution_time'].value.toInstant().toEpochMilli() -
  doc['detection_time'].value.toInstant().toEpochMilli()) / 3600000.0);
}
```

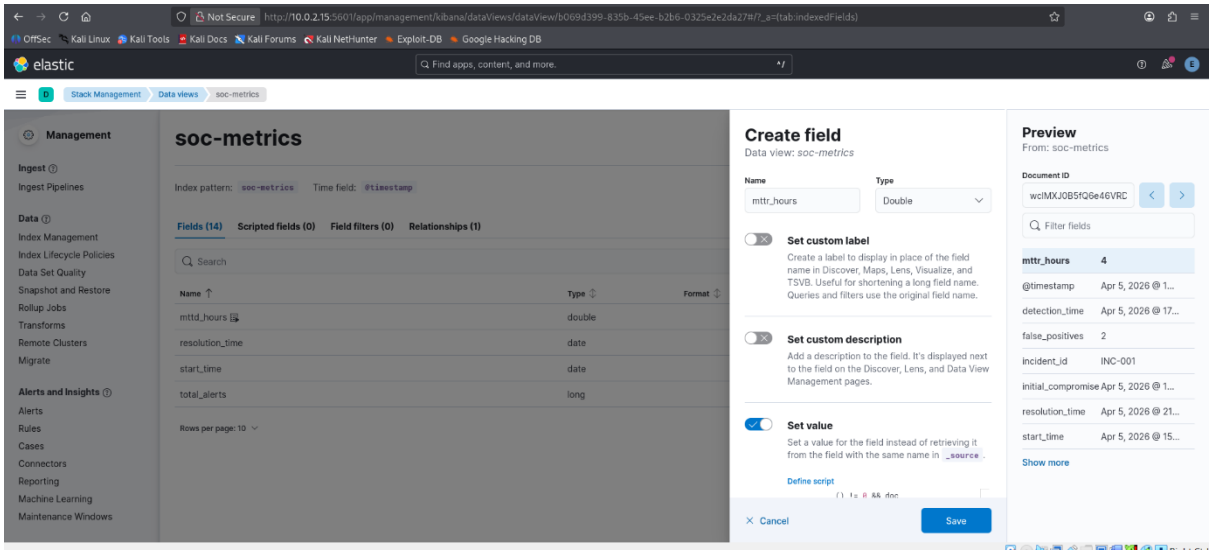

Figure 11: creating mttr_hours field

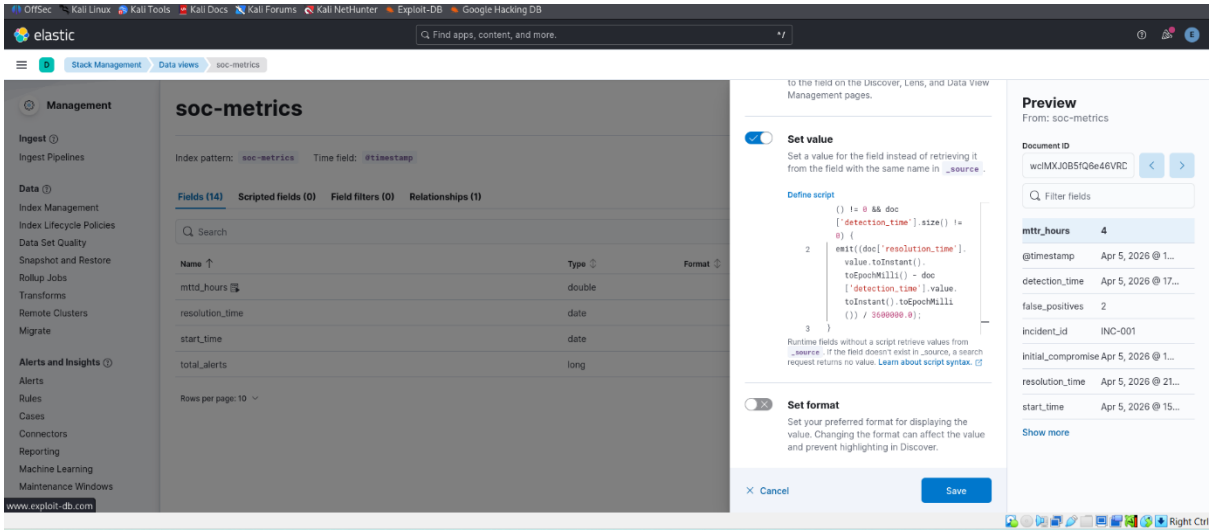

Figure 12: adding script to mttr_hours

It calculates MTTR and Then converts it into hours.

Field 3 – Dwell Time

Name: dwell_time_hours

Type: double

Script:

```
if (doc['detection_time'].size() != 0 && doc['initial_compromise'].size() != 0)
{
emit((doc['detection_time'].value.toInstant().toEpochMilli() -
doc['initial_compromise'].value.toInstant().toEpochMilli()) / 3600000.0);
}
```

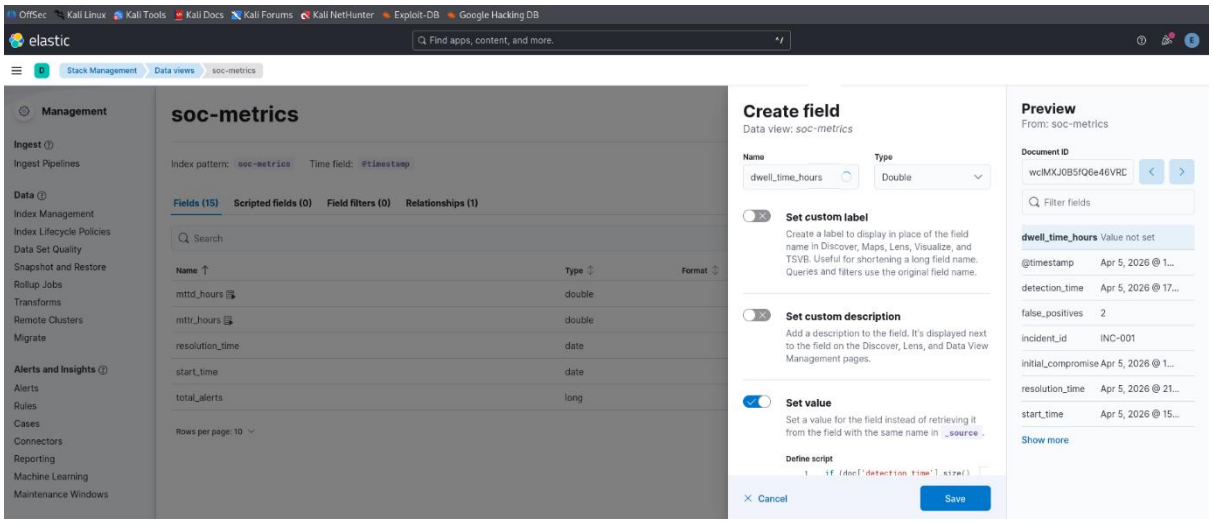

Figure 13: creating dwell_time_hours field

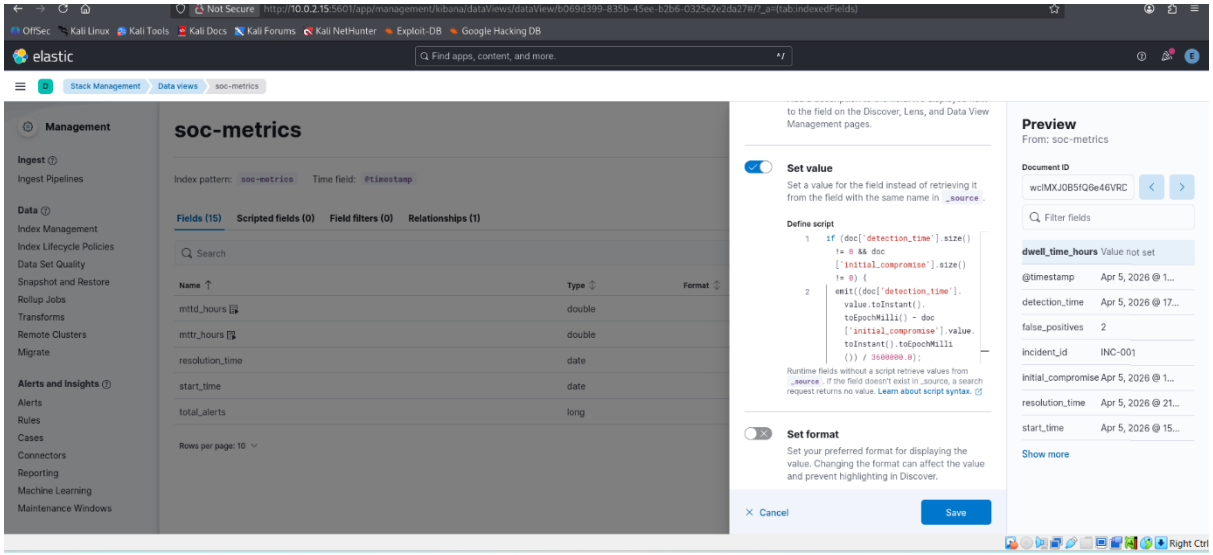

Figure 14: adding script to dwell_time_hours field

Field 4 – False Positive Rate:

Name: false_positive_rate

Type: double

Script:

```
if (doc['false_positives'].size() != 0 && doc['total_alerts'].size() != 0 &&
doc['total_alerts'].value != 0) {
 emit((doc['false_positives'].value * 100.0) / doc['total_alerts'].value);
}
```

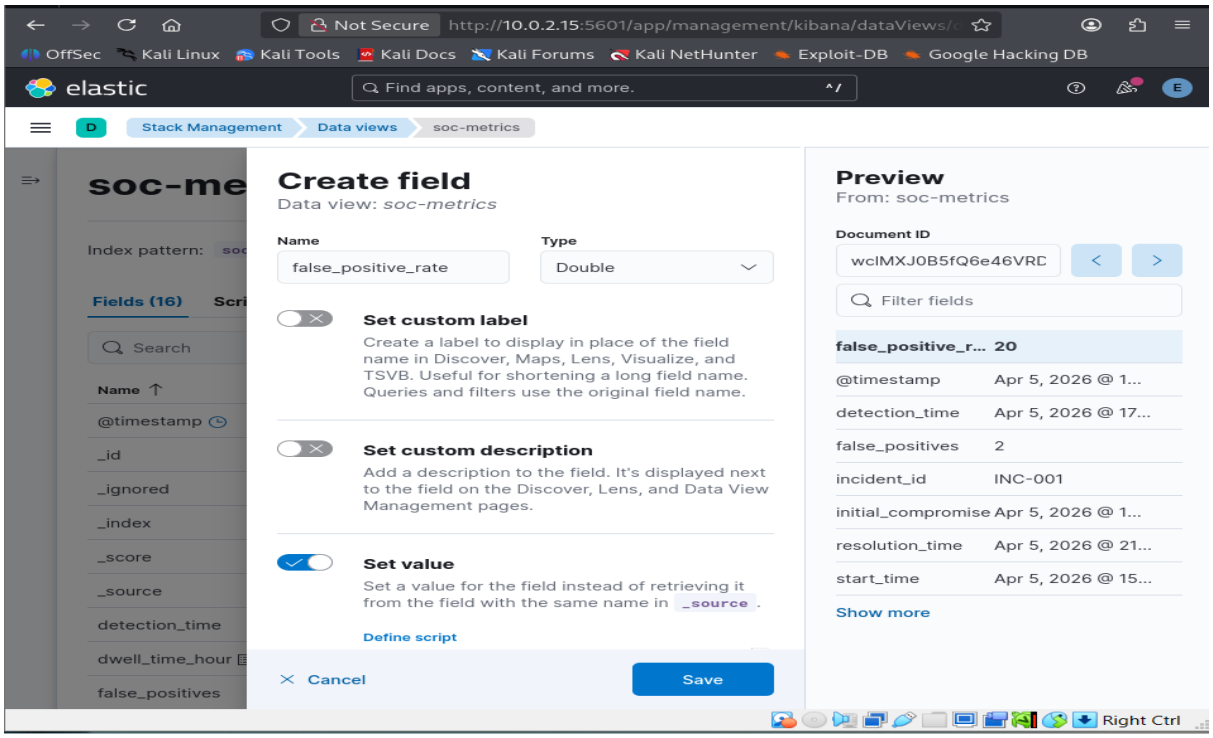

Figure 15: creating false_positive_rate field

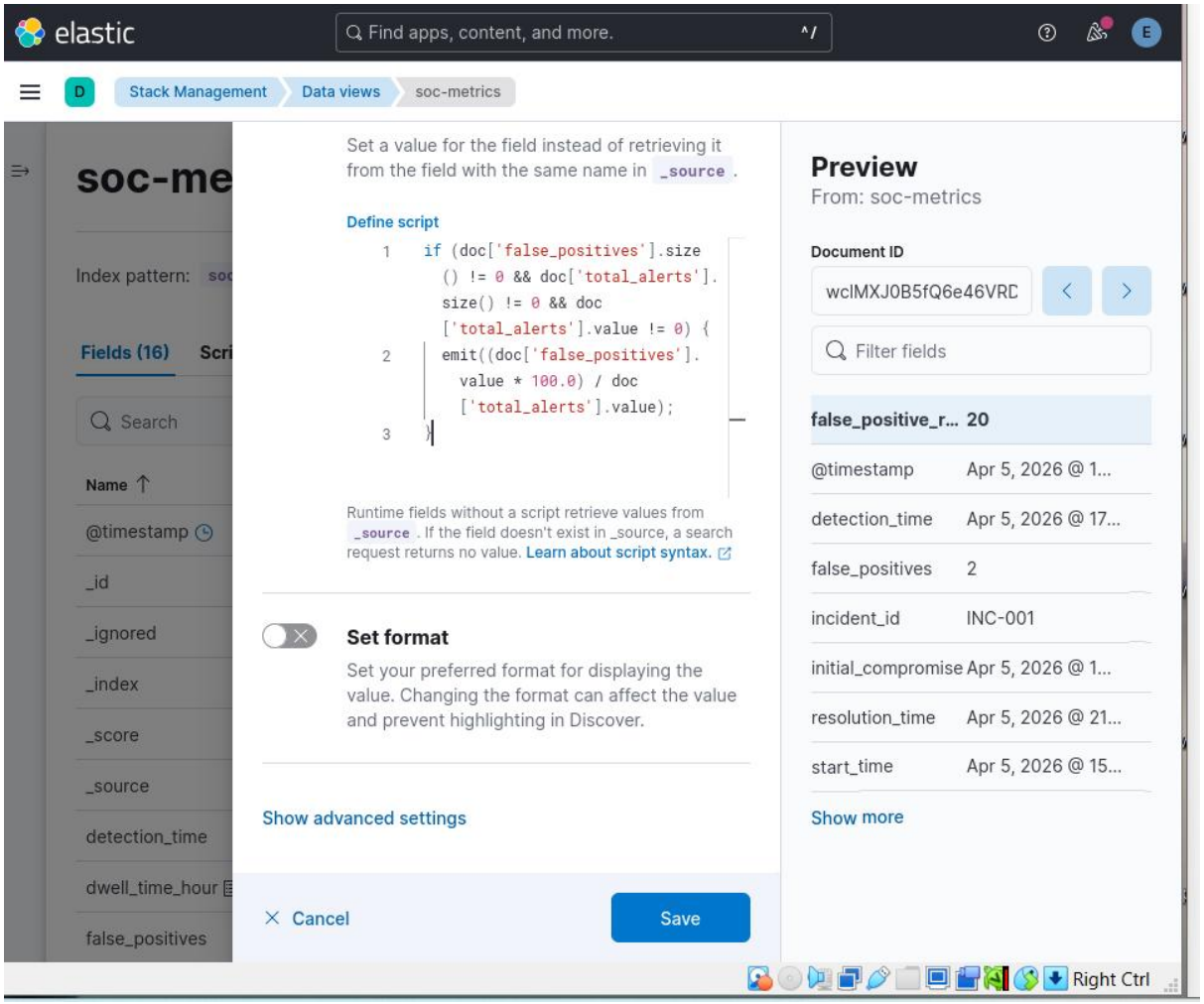

Figure 16: adding script to false_positive_rate field

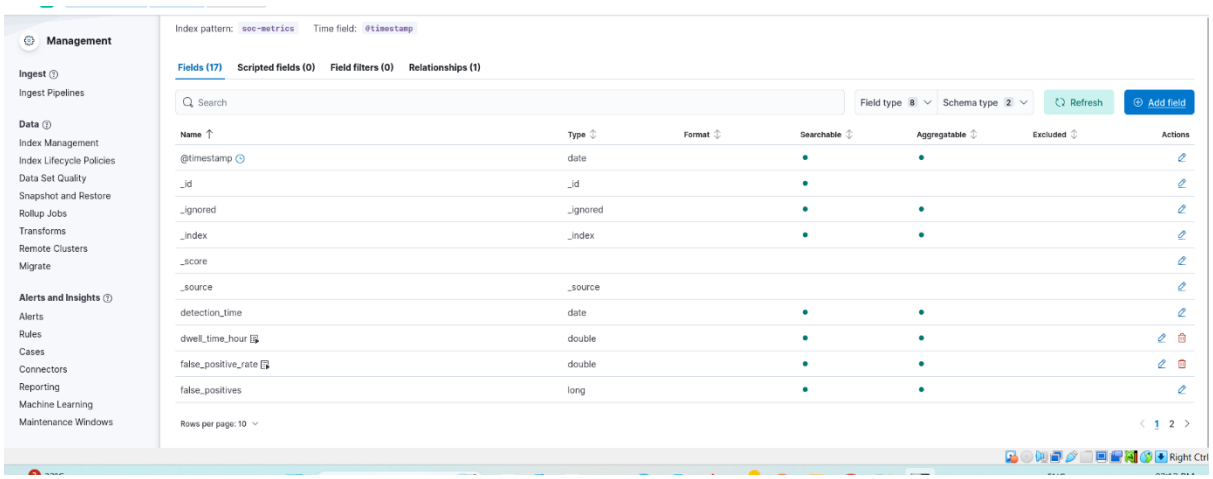

Figure 17: soc-metrics data view

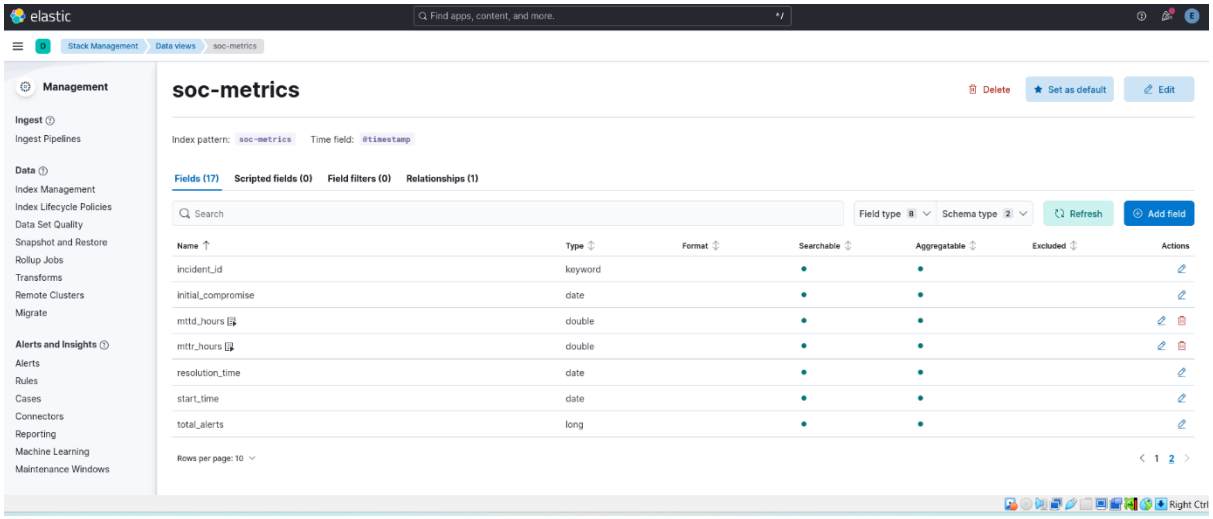

Figure 18: soc-metrics data view

5) Build the Dashboard in Elastic Security:

Goto analytics → dashboard → then click to create dashboard → after that add visualization

Visualization 1: Average MTTD

- Click Create visualization
- Select index: soc-metrics
- Drag mttd_hours
- Set metric to Average
- Title it:

  Average MTTD

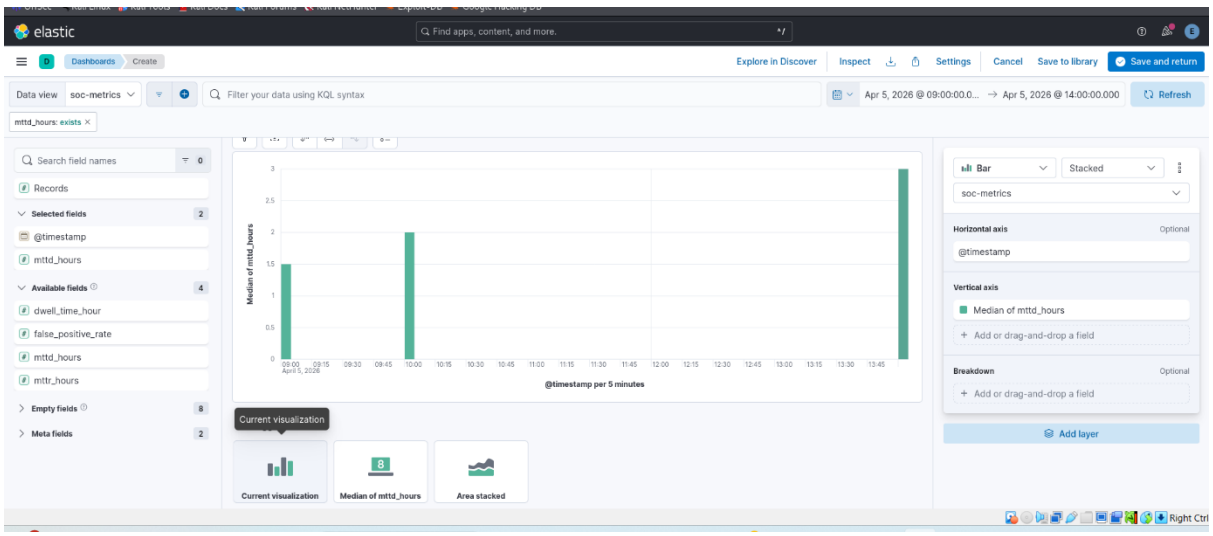

Figure 19: Bar Chart Visualization of MTTD

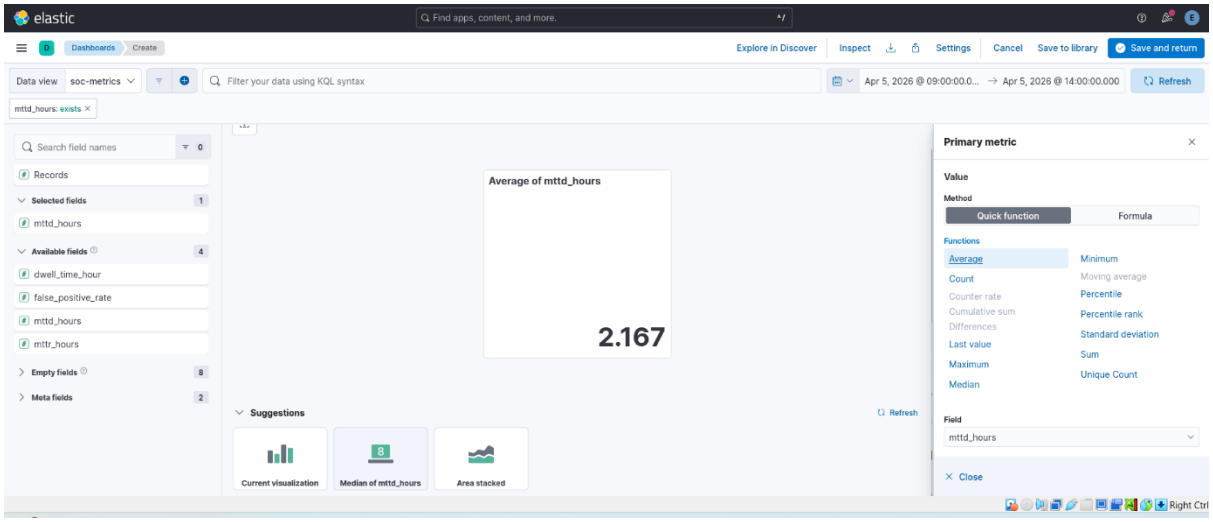

Figure 20: Average MTTD Metric Visualization

Average MTTD is 2.167 hours

Visualization 2 — Average MTTR
Repeat same steps as above and

Field: mttr_hours
Metric: Average
Title: Average MTTR

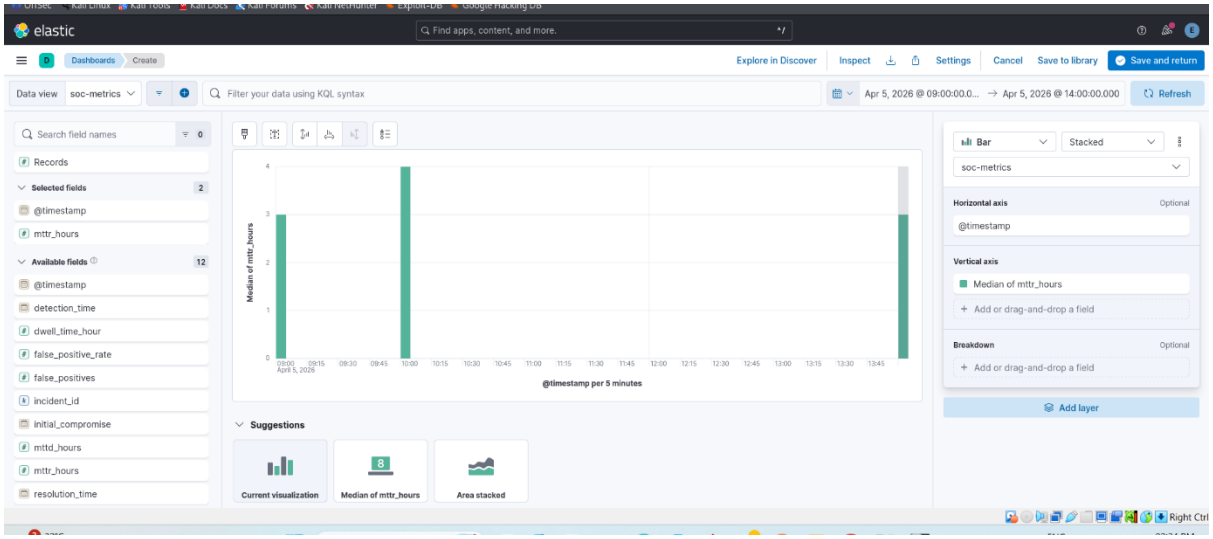

Figure 21: Bar Chart Visualization of MTTR

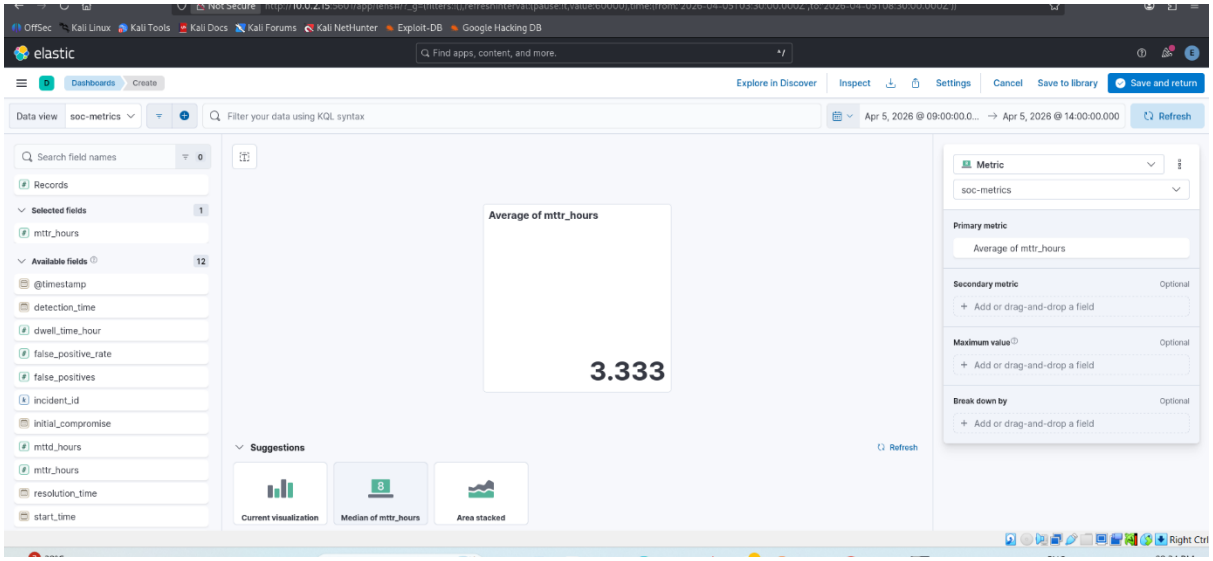

Figure 22: Average MTTR Metric Visualization

Average MTTR is 3.333 hours

Visualization 3 — Average False Positive Rate

Field: false_positive_rate

Metric: Average

Title: False Positive Rate

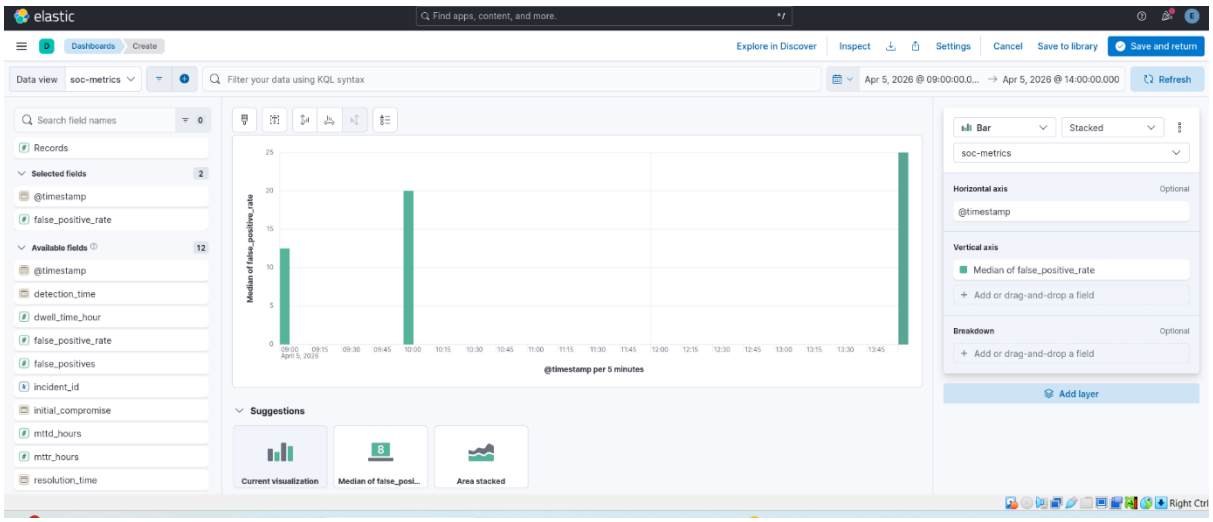

Figure 23: Bar Chart Visualization of false positive rate

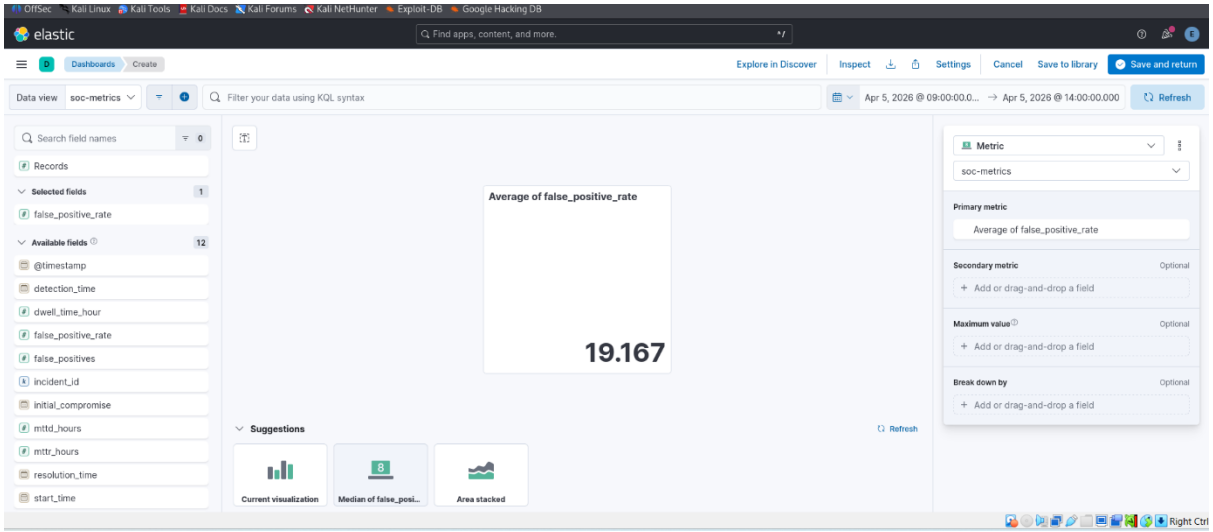

Figure 24: Average false positive rate Metric Visualization

Average false positive rate is 19.167 %

Visualization 4 — Dwell Time per Incident

- Create another visualization
- Use Bar chart
- X-axis = incident_id
- Y-axis = dwell_time_hours
- Title: Dwell Time by Incident

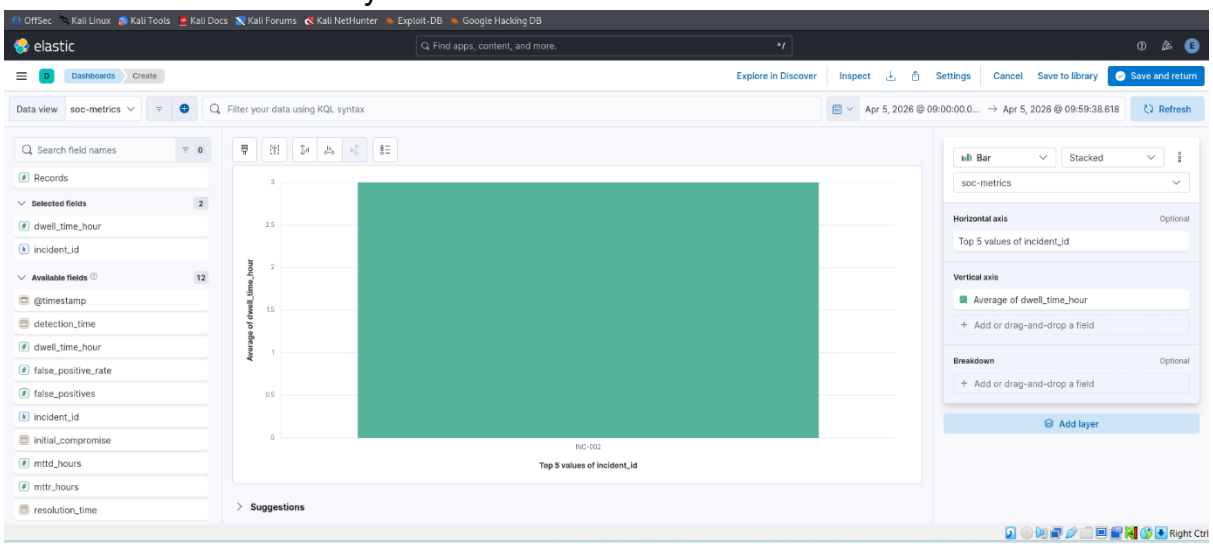

Figure 25 : above the screenshot shows Incident INC-002 remained undetected for about 3 hours before it was identified.

Visualization 5 — Alerts vs False Positives

- Use a Bar chart
- X-axis = incident_id
- Y-axis = total_alerts
- Add another metric = false_positives
- Title: Total Alerts vs False Positives

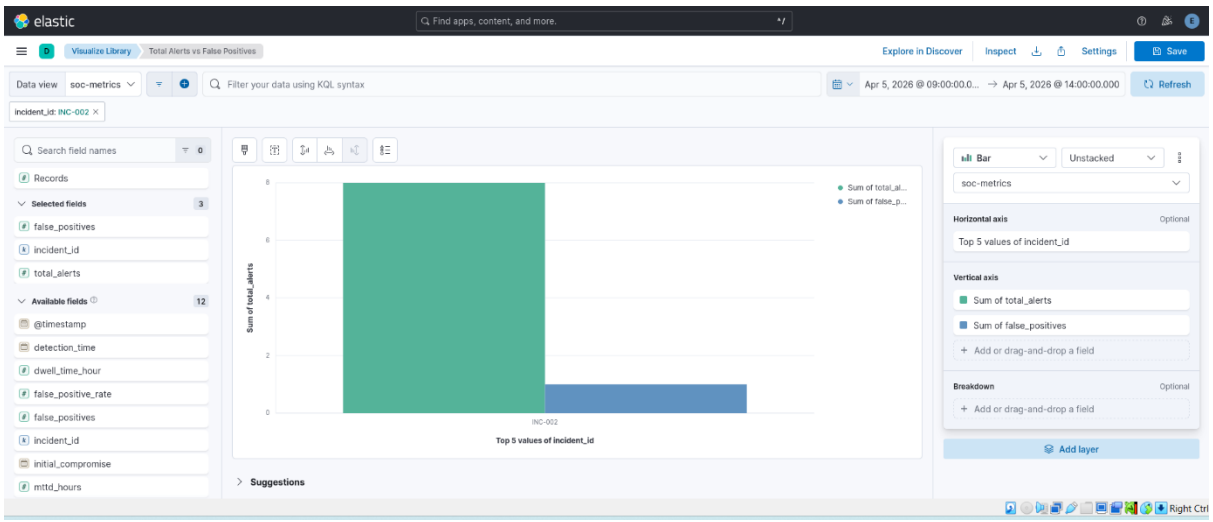

Figure 26 : above the screenshot shows there are sum of total alerts is 8 and sum of false positive is 1.

Dash board : SOC Security Metrics Dashboard

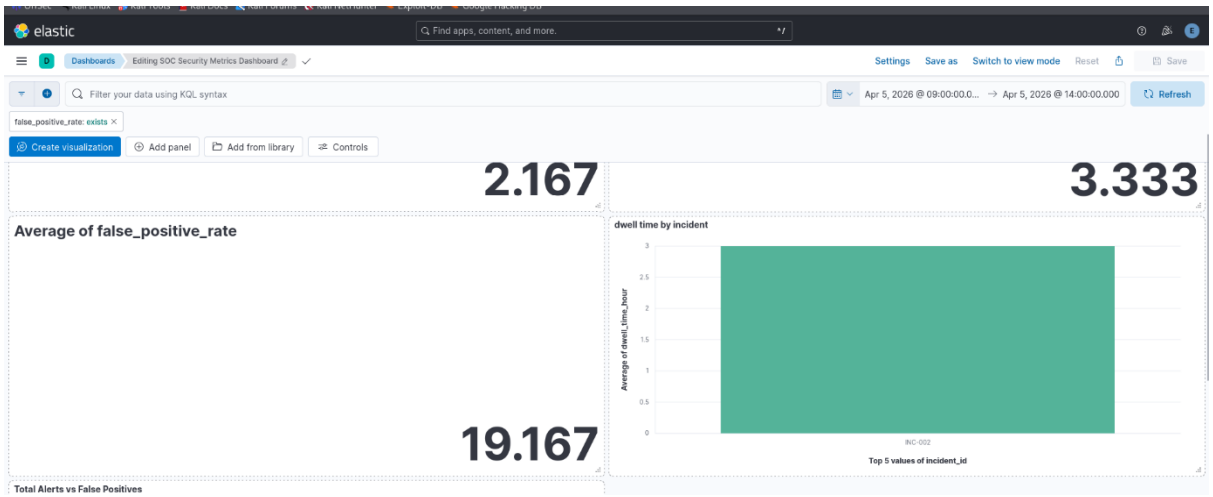

Figure 27

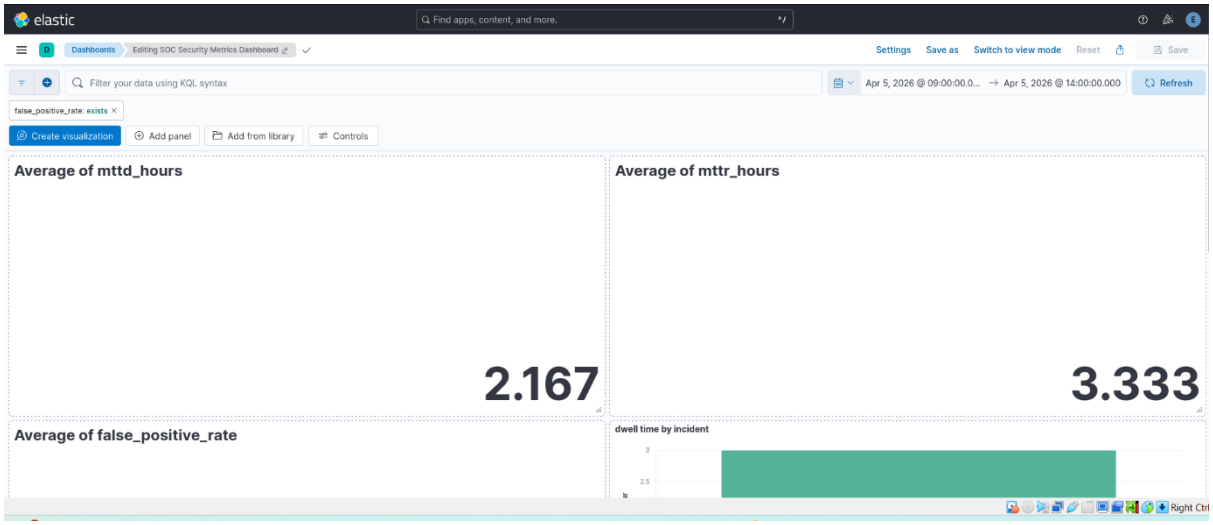

Figure 28

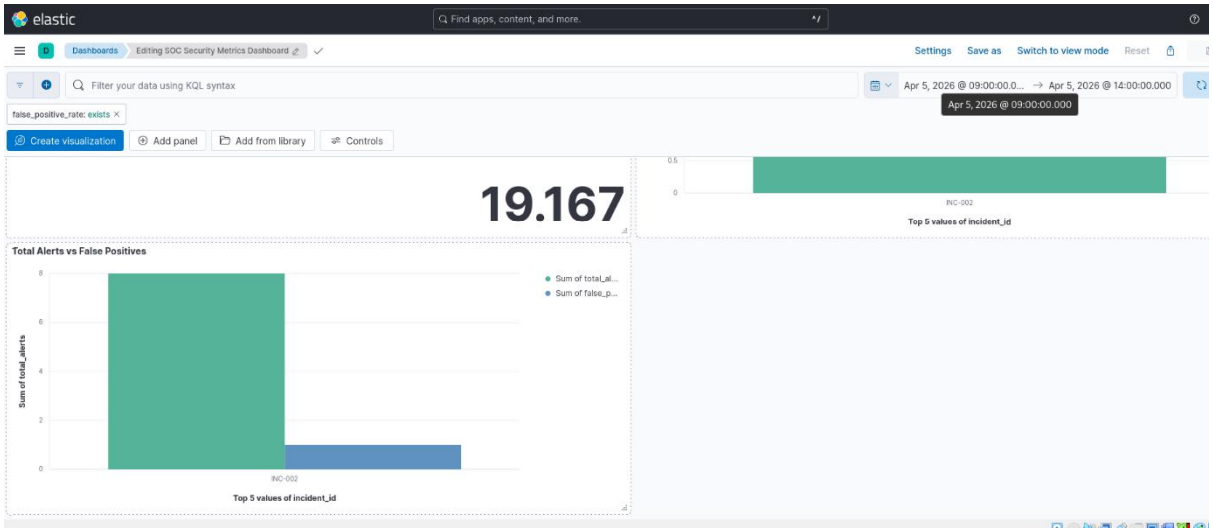

Figure 29

Figure 27,28,29 is SOC Security Metrics Dashboard.

The dashboard displays the average false positive rate, which helps understand how many alerts are incorrect. It also shows MTTD and MTTR, indicating how quickly incidents are detected and resolved. The dwell time shows how long a threat stays in the system before detection. It compares total alerts with false positives to check alert accuracy.The dashboard helps analysts improve detection rules, reduce false alerts, and respond to incidents faster, improving the security of the system.